



PF32 QUICK START GUIDE

v1.5.14

WARRANTY AND DISCLAIMERS

Use of this product and the associated software implies acceptance of the Photon Force terms of use.

Bare electronic components are susceptible to electrostatic discharge (ESD). Failure to employ good practice in handling the hardware, or to observe and comply with the warnings and handling precautions stated in this guide will void the product warranty.

This product may not be used in military, aerospace, medical or other safety critical applications without the express written permission of Photon Force Ltd. Any such use is undertaken entirely at the customer's own risk.

Specifications are subject to change without notice.

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WELCOME

Thank you for choosing to purchase a PF32 system from Photon Force. We look forward to supporting your research and hearing about your discoveries. Please do stay in touch!

We've put together the following quick start guide to help you get up to speed with your new camera. If you have not yet already downloaded the software and documentation for your PF32 camera, this available online at <https://www.photon-force.com/setup/>

Photon Force supports a number of ways to use your PF32 camera, all built around our camera driver. A Java demo application is provided, which can be used to quickly test the camera, view a video stream, capture histograms, and explore the camera settings. The Java application is discussed later in this guide. A MATLAB® wrapper is provided for scientific use, allowing the camera to be operated and data acquired from within the MATLAB® environment. Demo functions are provided for acquiring both live video for display, and high speed streaming of data for analysis. The MATLAB® wrapper is discussed in a dedicated guide, available from the setup link above. Finally, you may use the Photon Force API to communicate with the PF32 camera from your own software. Please consult the API guide provided via the setup page above.

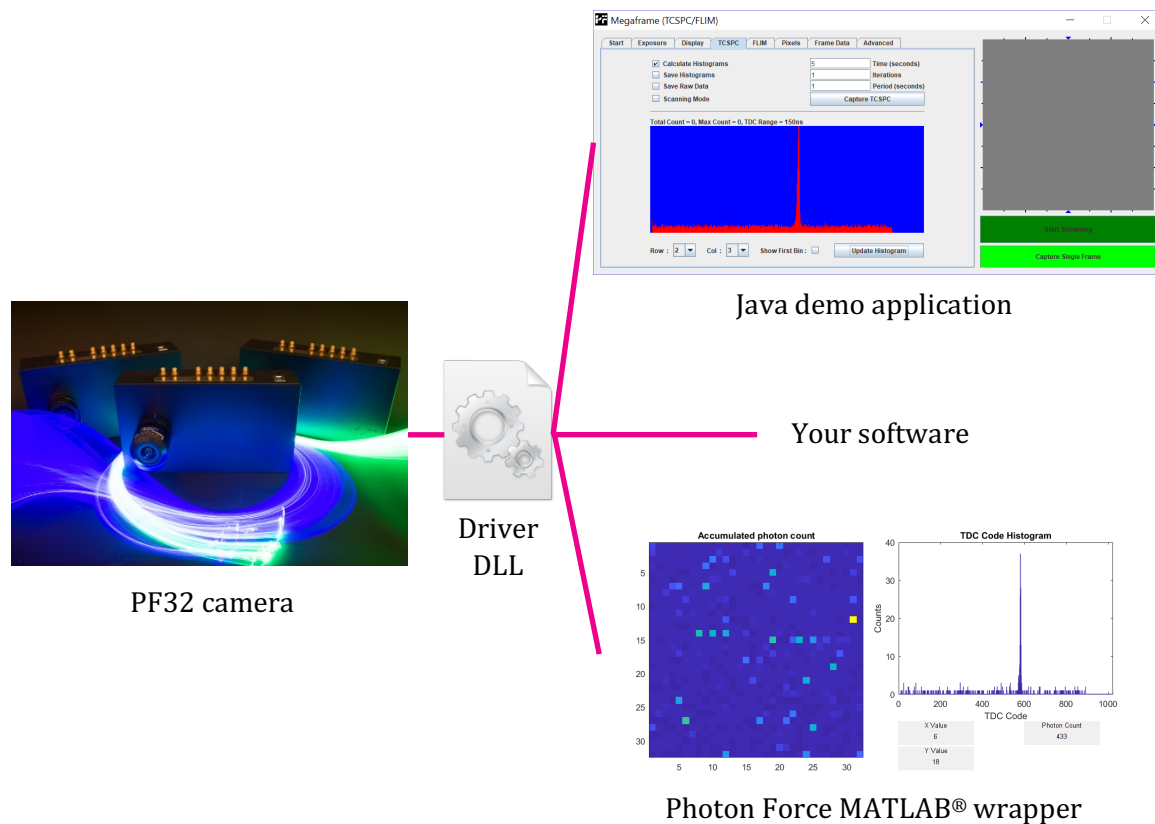


Illustration of software components.

SOFTWARE INSTALLATION – MICROSOFT WINDOWS

The PF32 application and USB driver are designed for 64-bit Windows 10, but are also compatible with older versions of Windows. 32-bit Windows and Linux versions are available upon request.

Software installation is performed in two steps:

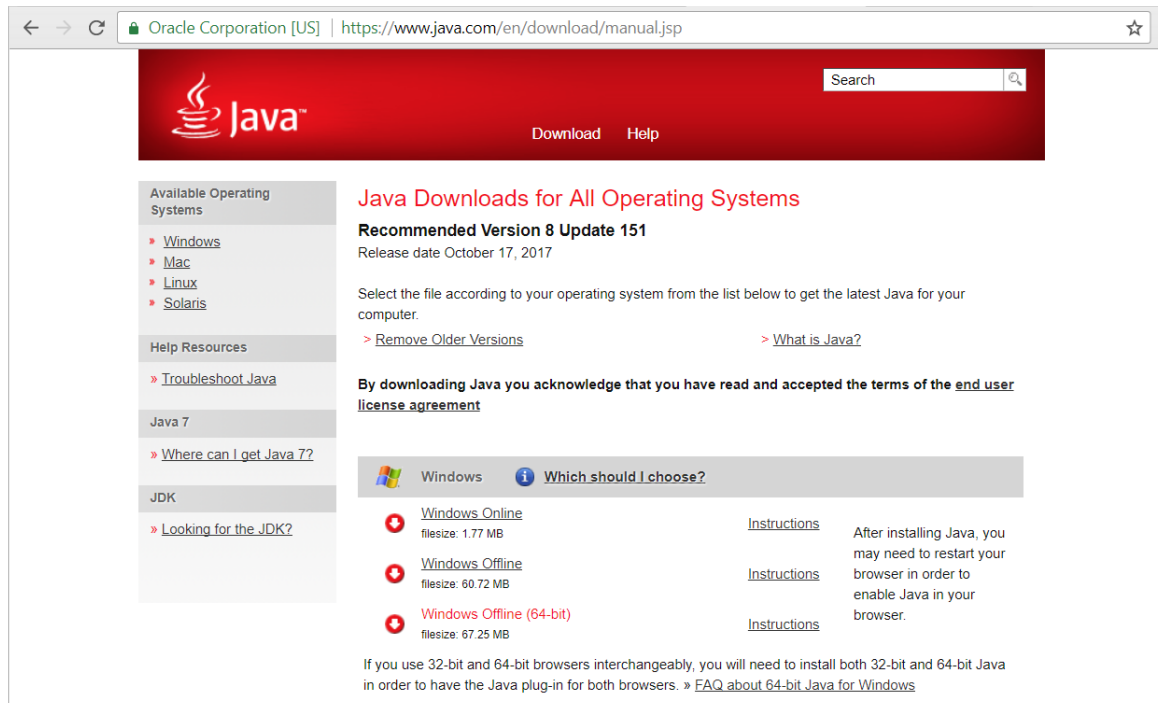
1. Obtain up-to-date Java Runtime Environment (JRE).
2. Install PF32 driver and demo application.

JAVA RUNTIME ENVIRONMENT INSTALLATION

The PF32 Java application requires the Java Runtime Environment (JRE). If not installed already, please download the JRE from the Oracle Java web site:

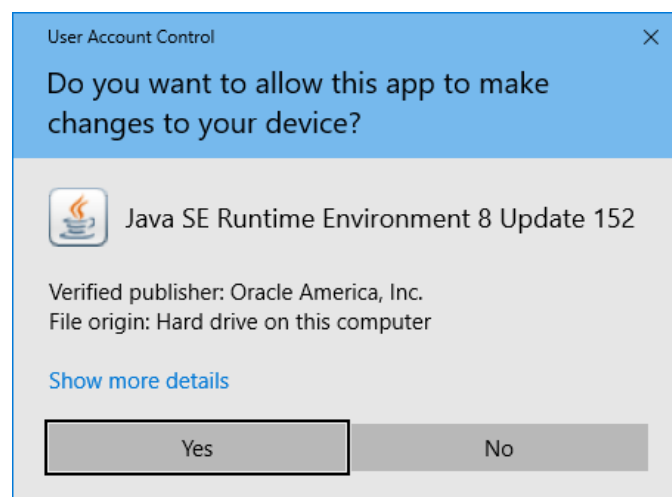
<https://www.java.com/en/download/manual.jsp>

To install the Oracle Java Runtime Environment, download and run the Windows Offline (64-bit) installer.



The installation of the JRE requires the following steps:

1. Run the installer and provide the necessary administrator rights to allow the installation.



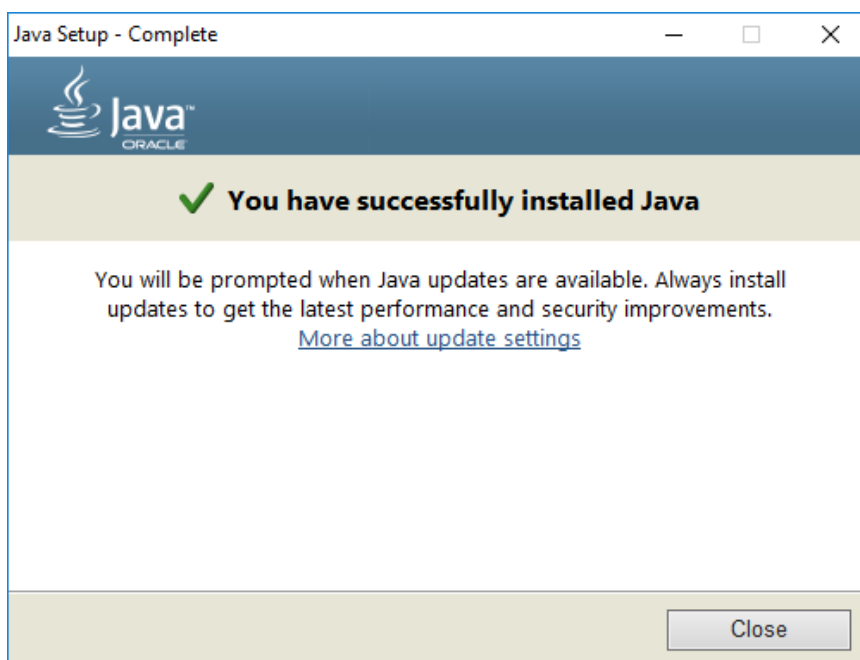
2. Accept the license agreement.



3. Wait for the installer to finish.



4. Close the installer and proceed to the installation of the PF32 software package.

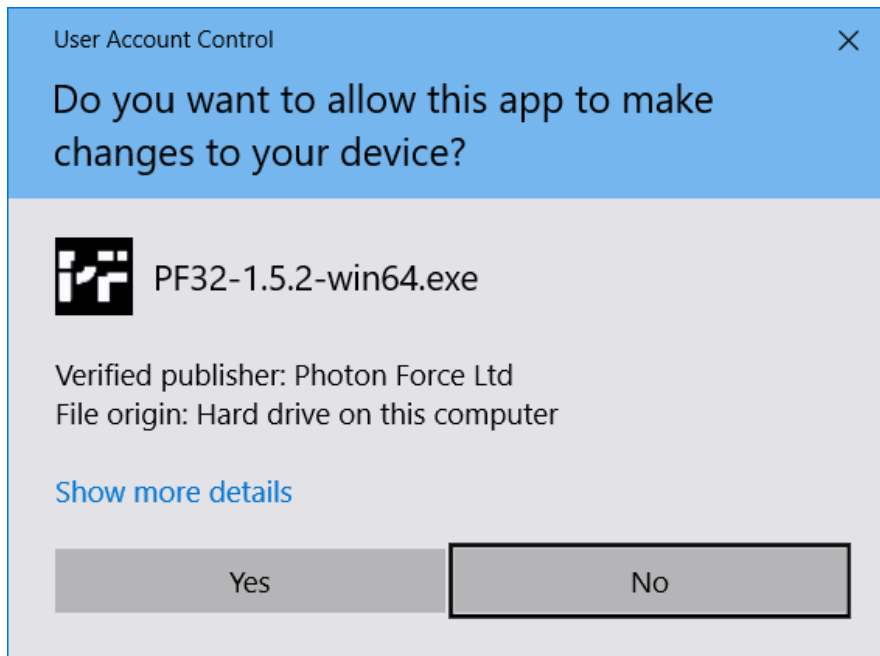


PF32 SOFTWARE PACKAGE INSTALLATION

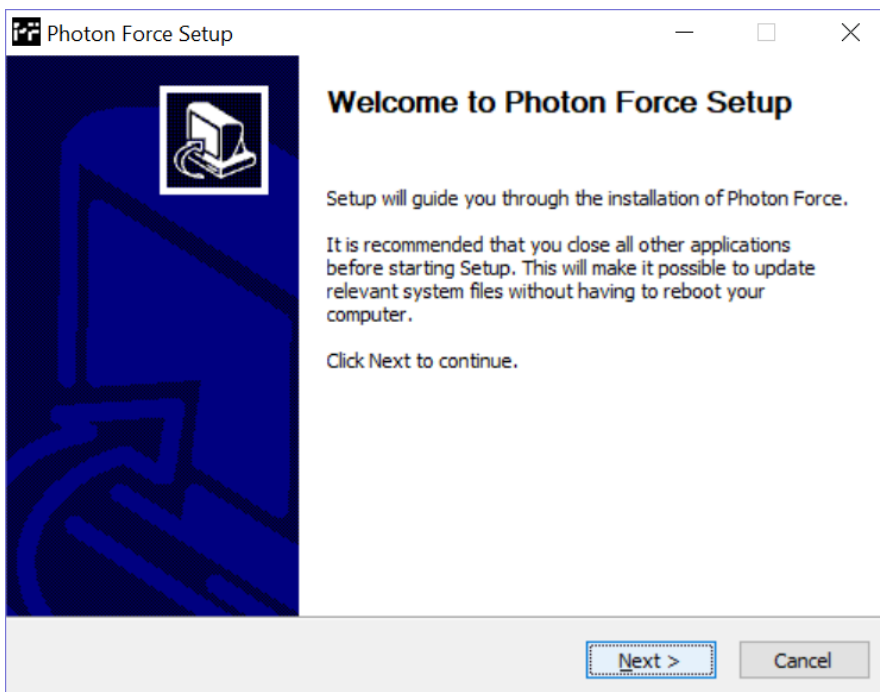
Regular updates for the USB driver and Java demo application are offered to Photon Force customers.

Run the PF32 installer. The latest version 1.5.6 for Windows is called:

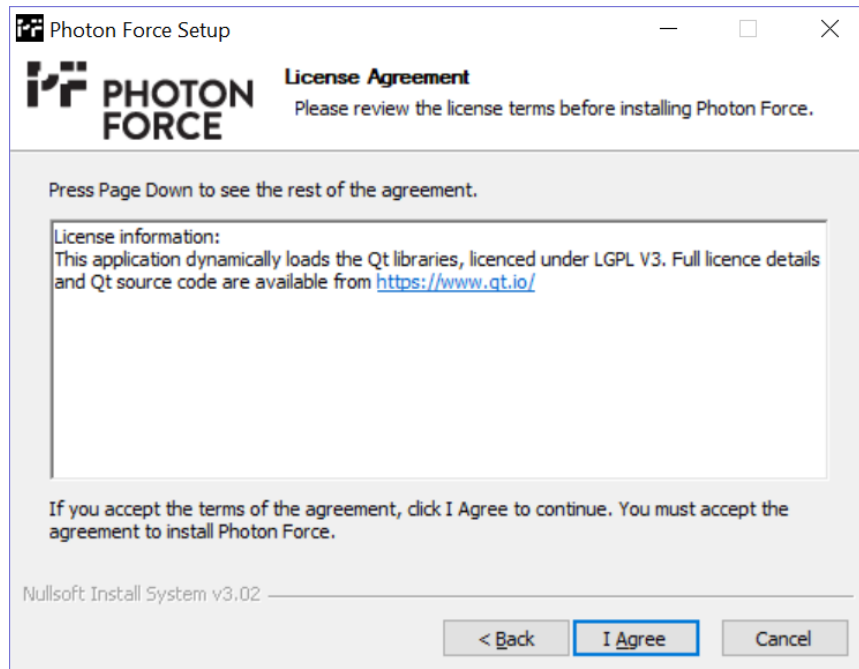
1. The installation may require Administrator permission. Press "Yes" when the following window appears and provide the Administrator password if required.



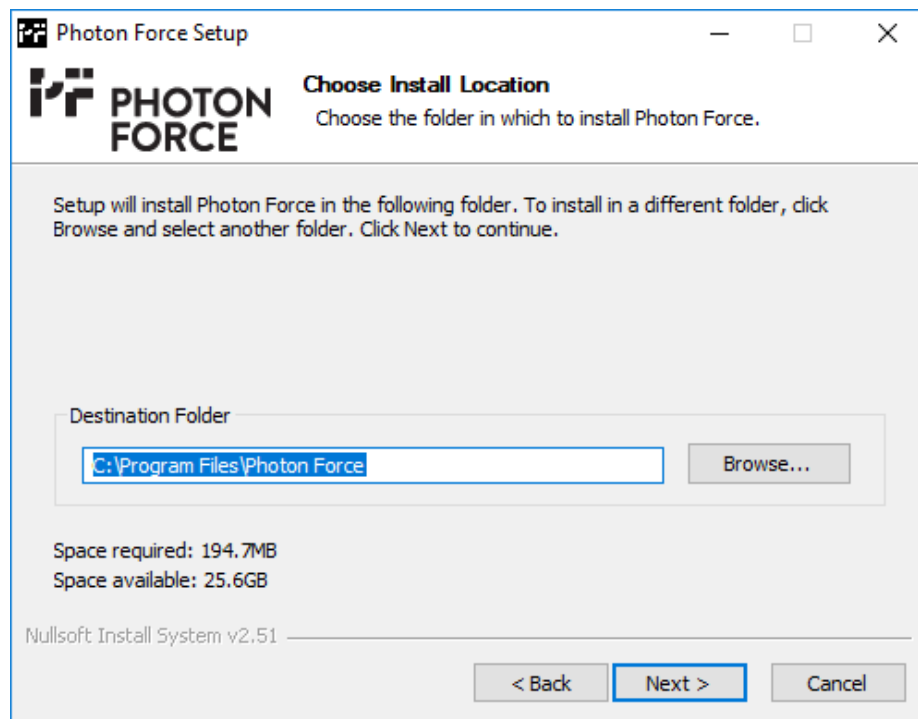
2. Start the installation.



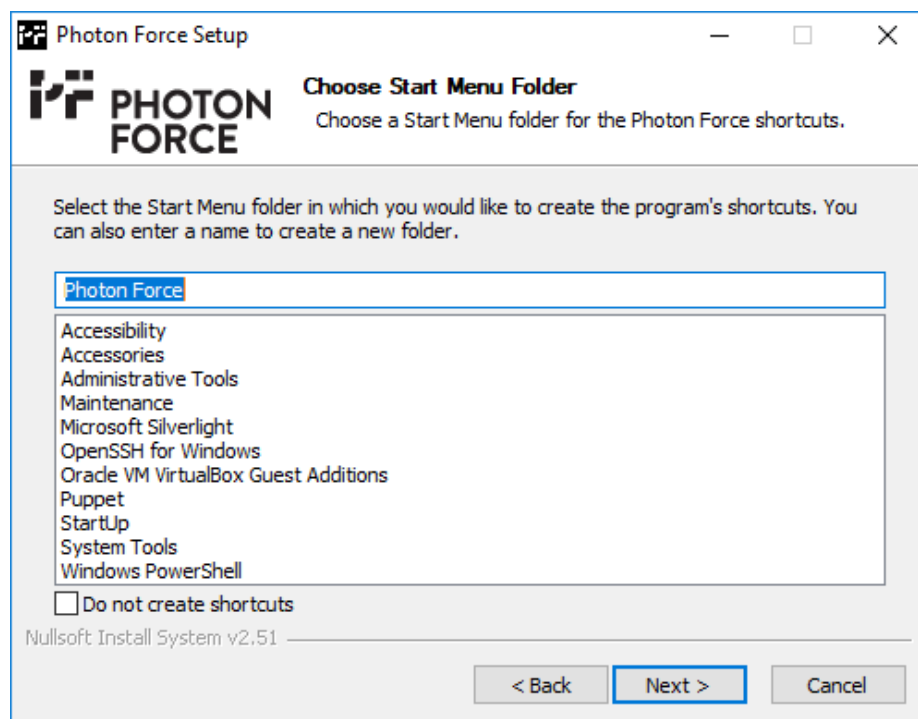
3. Accept the license agreement.



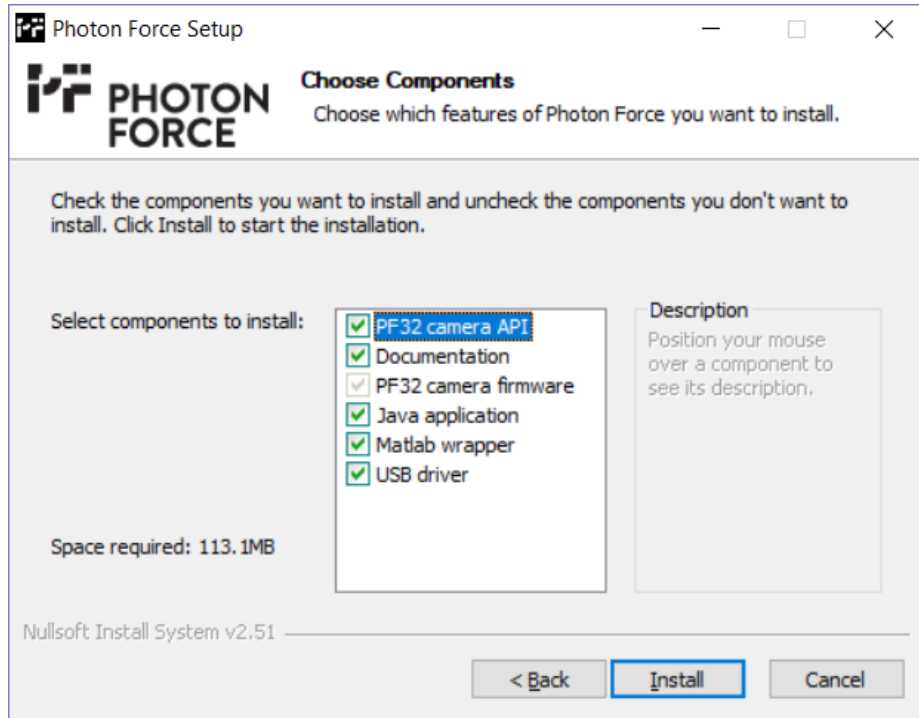
4. Choose the installation folder.



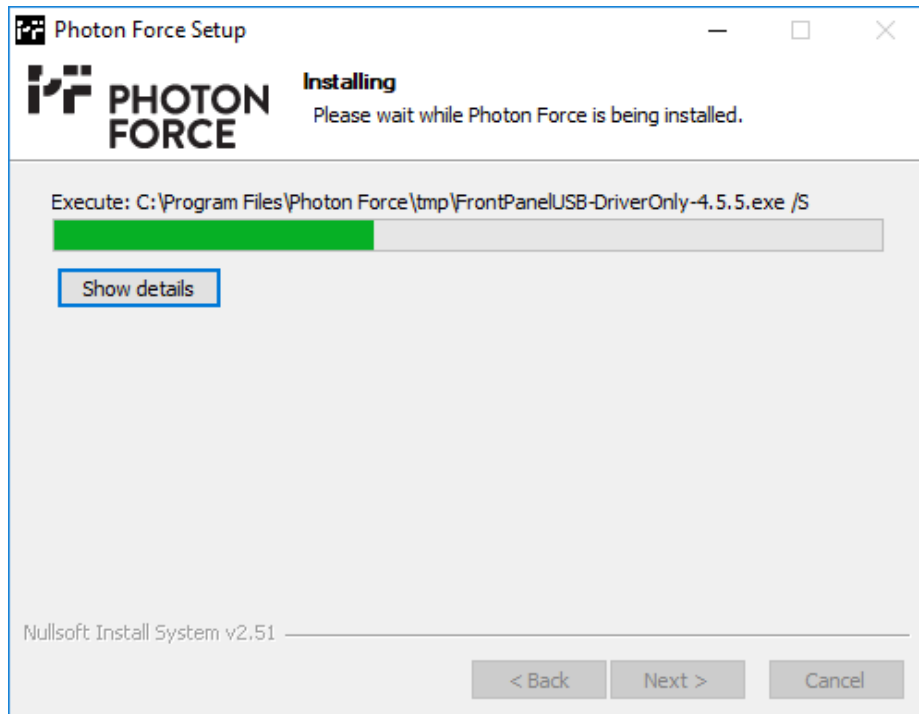
5. Choose the start menu folder in which Photon Force shortcuts will be created.



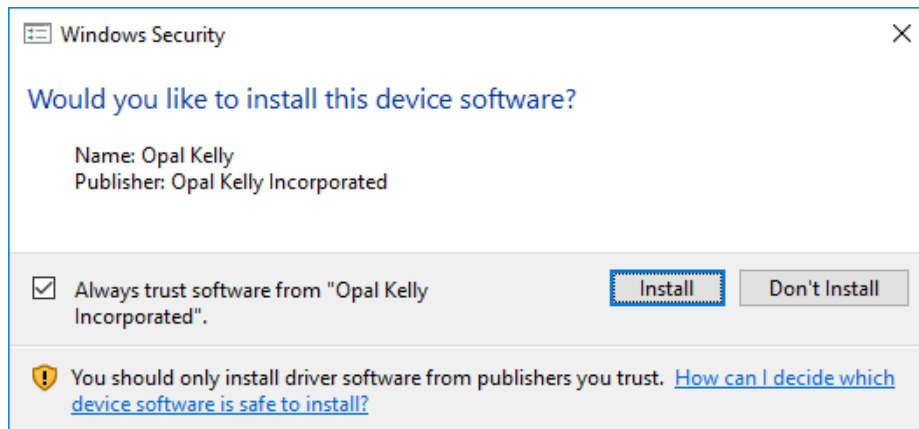
6. Choose the components to install. The *PF32 camera API* is the software interface to the camera. The *Java application* is a standalone demo application and GUI to run the camera. The *USB driver* is required for operating system communication with the camera.



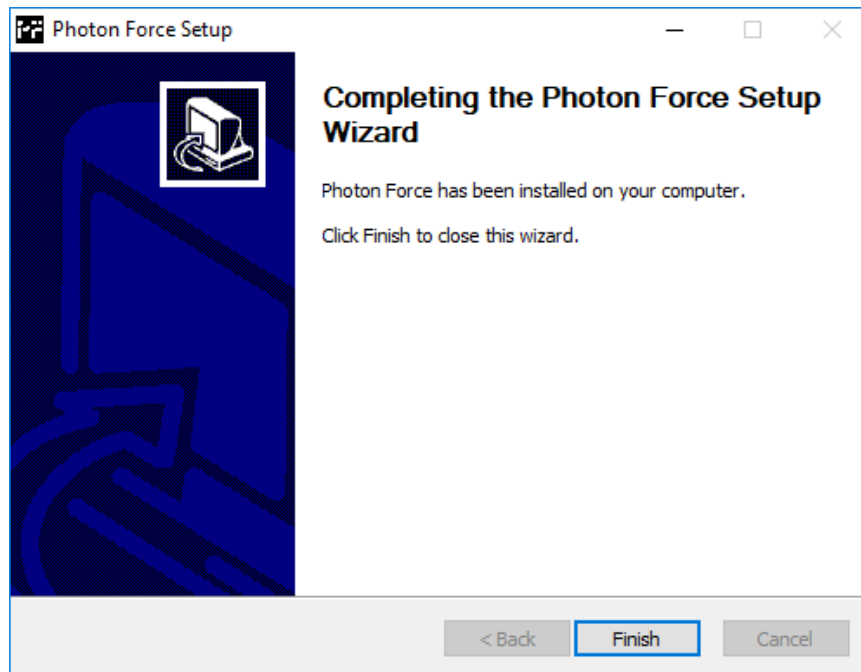
7. Wait for the installer to finish.



8. Provide permission to install the driver.



9. Finish the installer.



PF32 JAVA DEMO APPLICATION FIRST RUN

USB3 and Power Connection

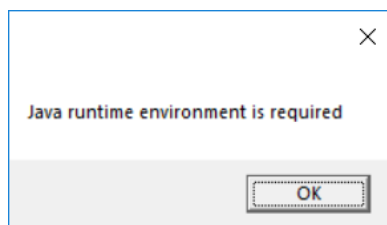
Included within the PF32 carry-case are the 5V power supply (PSU) and the USB3 cable. It is crucial that no other PSU is used since an incorrect voltage, polarity, or current rating may result in incorrect operation or irreversible damage of the PF32.

Please connect the USB3 cable and PSU. Windows will then perform a driver installation the first time the board is connected.

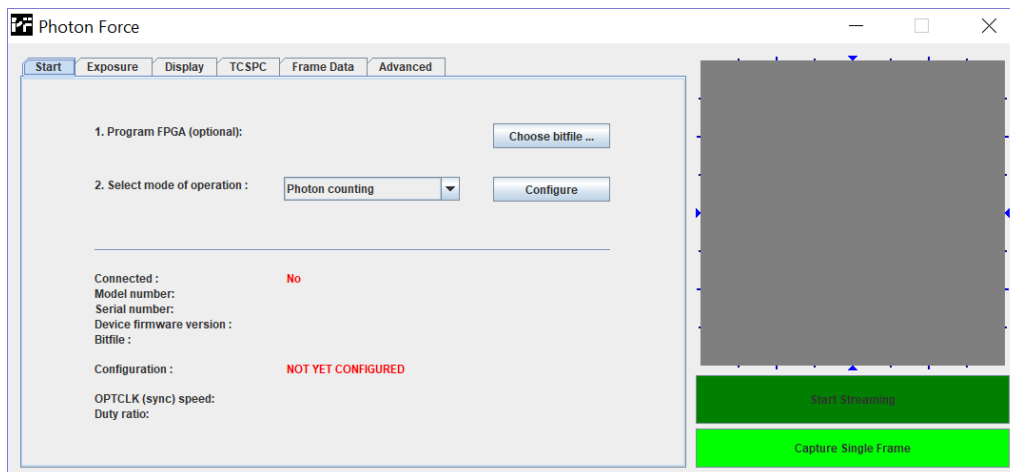
Start the *PF32 Java Win_x64* application.

If the program does not start at all, ensure that the Java Runtime Environment is installed first (see [here](#)).

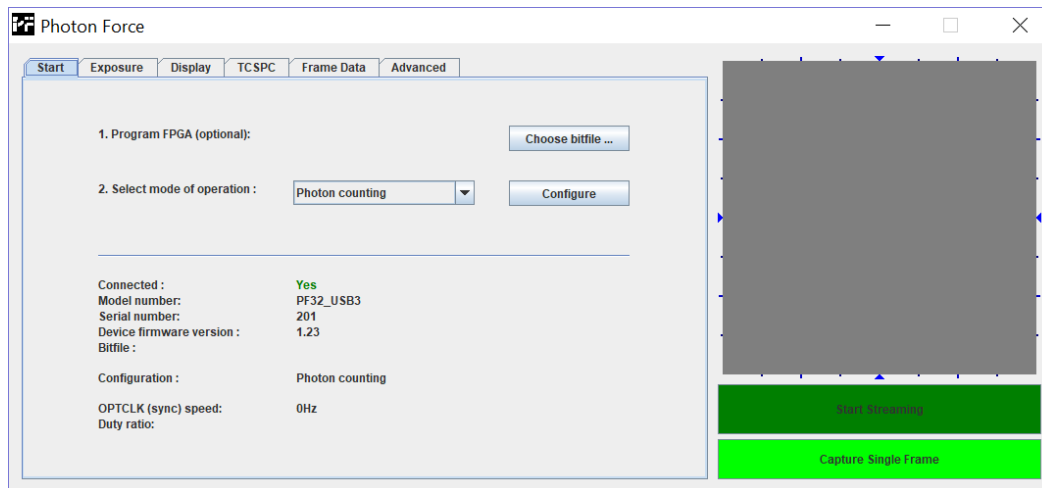
1. Note that if the Java Runtime Environment has not been installed, the following warning will be displayed. Please install a Java Runtime Environment as described above before continuing (see [Java Runtime Environment Installation](#)).



2. Once the application starts, the following window appears (if the camera isn't plugged in or powered on).



3. Ensure the PF32 system is connected and powered on and the window will update as shown below.



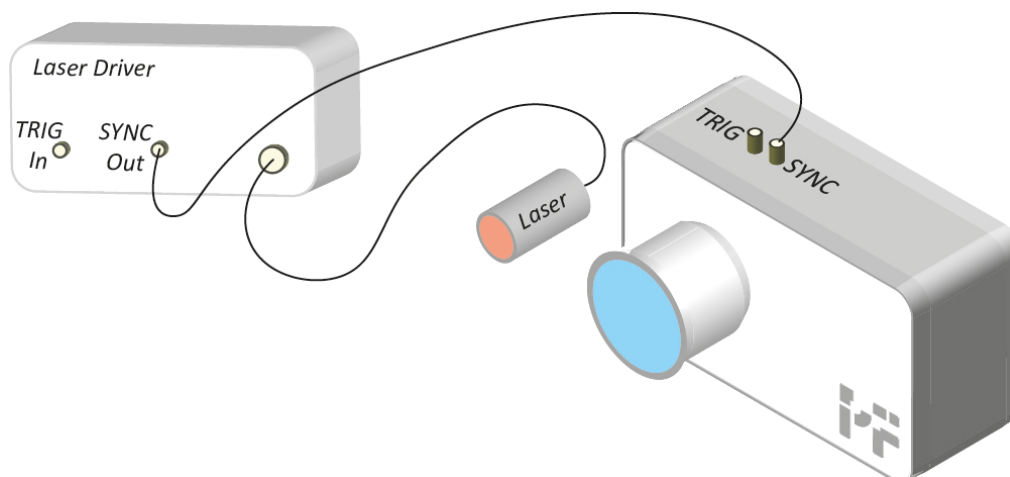
By default, the system is configured in a photon counting mode, which does not provide timing information. Settings may be changed by selecting the desired mode of operation from the drop-down menu and pressing configure, or by changing the appropriate settings in the control tabs.

At this point, in order to see an image within the software, you should attach a lens to the PF32. The PF32 has a CS-mount on the front. Should you wish you to use a C-mount lens, an appropriate spacer should be used. To use any other lens, adapters must be purchased separately.

CONNECTING THE CAMERA FOR TCSPC MODE

The instructions for [USB3 and Power Connection](#) should already have been followed.

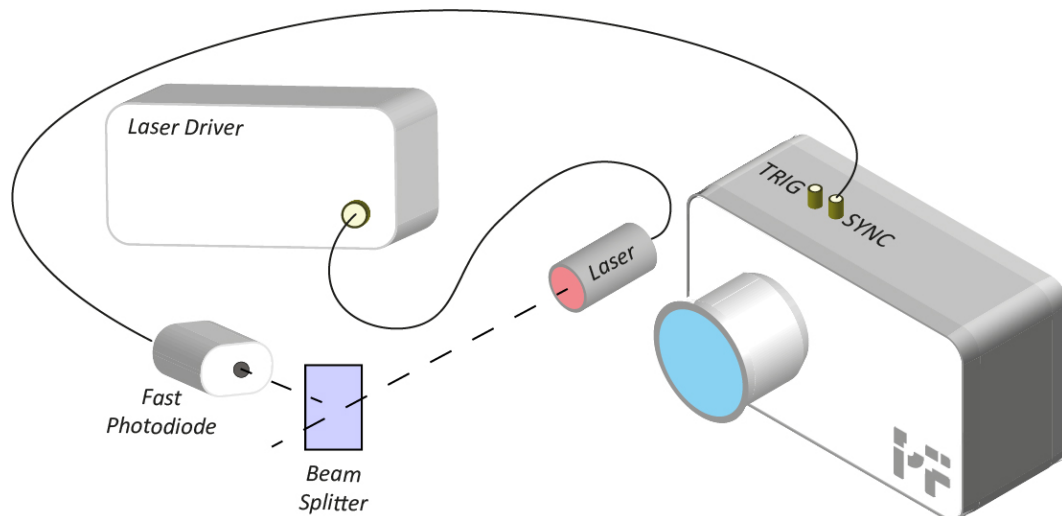
In order to synchronise your camera with a pulsed light source, there are a number of options available to the user. The most common of these is for the camera to be used in slave mode, whereby the timing reference is provided by an external source as shown in the example below:



PF32 Camera in slave TCSPC mode where the timing synchronisation is provided by the same clock as the light source.

In this case, the laser driver provides an electrical synchronisation reference pulse to the camera at the same time as the laser sends a pulse of light (to the target/sample depending on the application). The camera, as standard, accepts +3.3V (measured with 50 Ω load) pulses as reference through the SMA connector labelled "SYNC", but can be configured to accept other signals (NIM, for example) upon request. Care should be taken to ensure that the limits for the synchronisation signal are not exceeded to prevent damage to the electronics and sensor. Note, in the above example, the laser driver can also be triggered by an external source. In this case, the external source could trigger both the PF32 and the laser, provided the electrical pulse is within the limits stated in the PF32 SYNC Input Configuration document (supplied separately according to the SYNC options chosen by the customer).

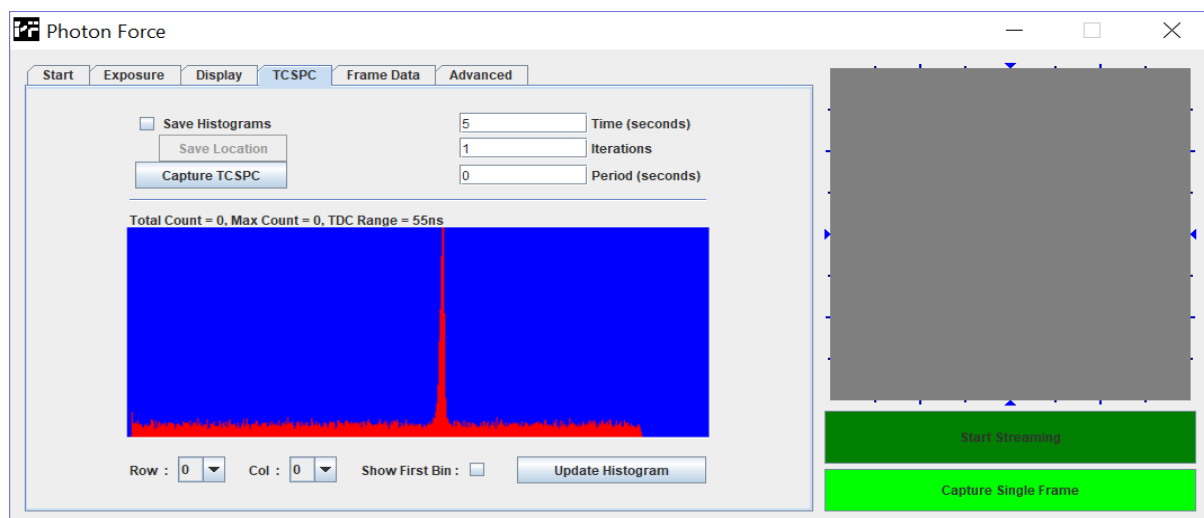
If the laser control unit does not include a synchronisation output, a portion of the outgoing beam can be split off (using an appropriate beam splitter) and directed on to a fast photodiode with circuitry that can provide an appropriate output signal. Photon Force can advise on the choice of trigger detector. Please contact us if you have any queries in this area.



PF32 camera in slave TCSPC mode where the timing synchronisation signal is provided by an optical detector (fast photodiode, for example) with an appropriate output.

With the SYNC signal connected to the camera, open the Java software. As before, program the FPGA by clicking on “Choose bitfile”, then select “PF32_USB3.bit”. Check that the sync rate displayed on the start tab matches the expected value. In the box “Select mode of operation” select “TCSPC (laser master)” then click “Configure”. Navigate to the TCSPC tab.

The simplest test to observe time-correlated data is to direct the pulsed light source, safely and diffusely, at a wall or scattering surface and image this surface with the camera. Within the TCSPC tab, click on “Capture TCSPC”. The default measurement time is 5 seconds, after which, the measurement will stop. Next, select which pixel’s histogram you wish to see by selecting a “Row” and “Col” value and click “Update Histogram”. Below is an example of a TCSPC histogram.

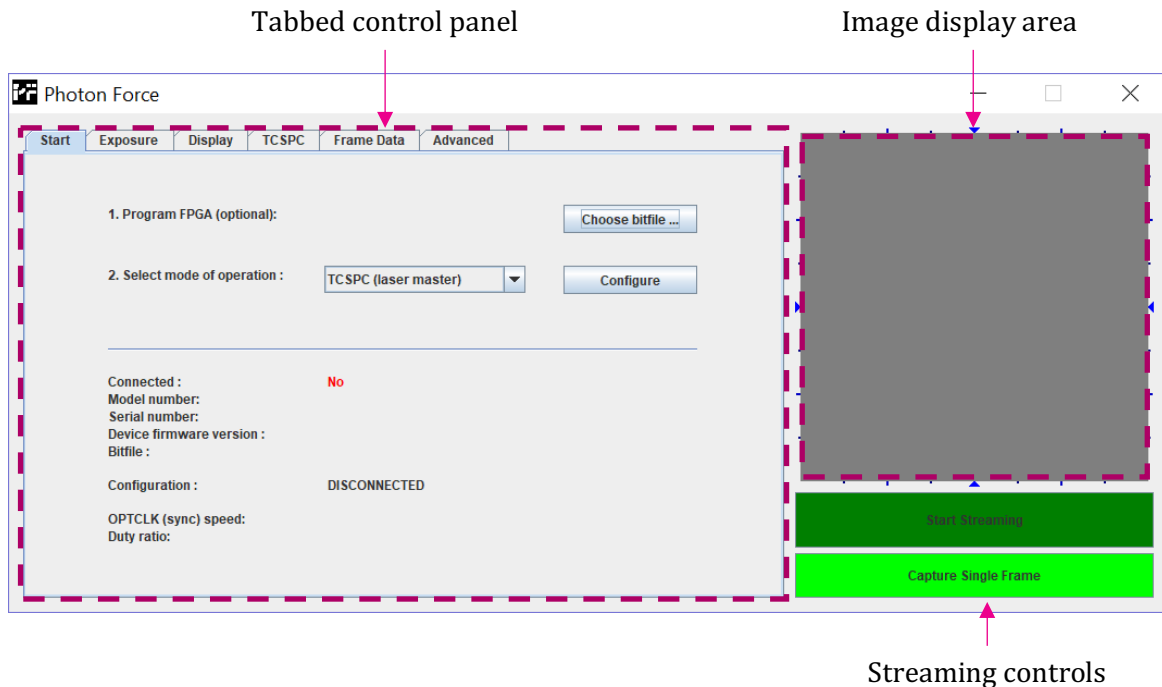


PF32 Java application displaying a captured histogram.

JAVA DEMO APPLICATION GUIDE

OVERVIEW

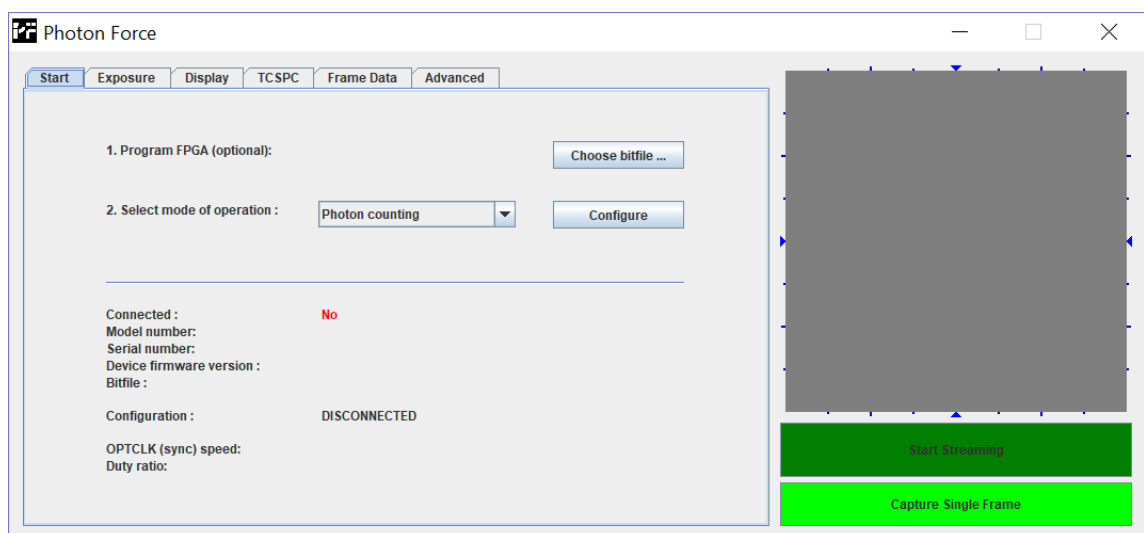
The PF32 Java demo application GUI is shown below. It consists of a tab-based control panel, an image display area, and streaming controls. These features are discussed in turn. Note that none of the features will be active until a PF32 camera has been connected, powered on, and configured using the start tab as described below.



Overview of the main PF32 Java demo application GUI features.

START TAB

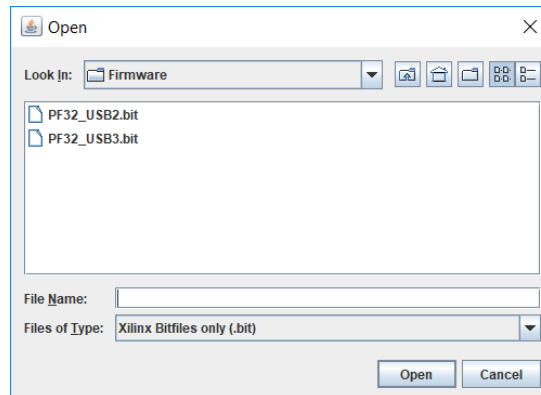
When the PF32 Java GUI is launched, it will initially display the Start tab, as shown below. The start tab displays status information, and provides controls to load the camera's firmware and set the operating mode.



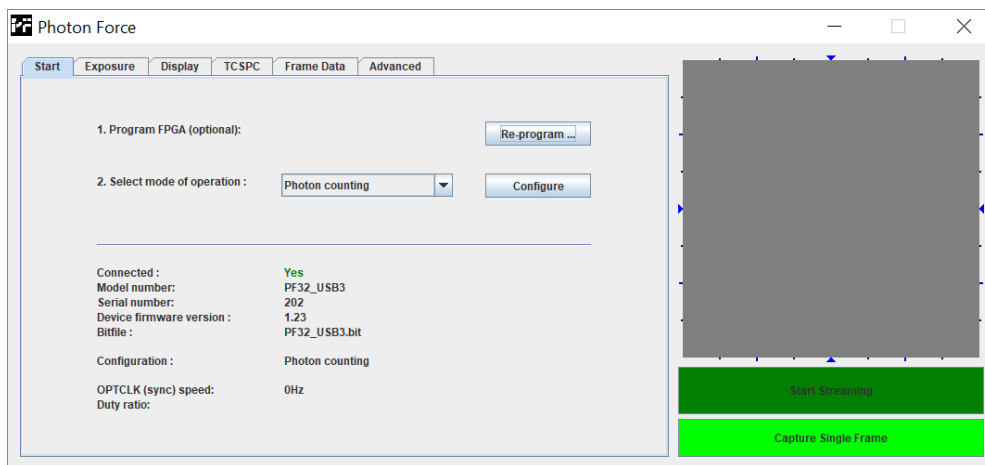
PF32 Java application Start tab, with no PF32 camera connected.

Note that none of the software features are active until a PF32 camera has been connected, and the firmware loaded. The default firmware will be loaded automatically whenever a PF32 camera

is connected and powered on. If you have been provided with custom firmware by Photon Force, press the “Choose bitfile” button, and select the appropriate firmware bit file for your PF32 camera (either USB2 or USB3) from the dialog as shown below.



The start tab should then update as follows:



PF32 Java application Start tab, with a camera connected and configured.

The following status information is provided in the start tab:

Item	Description
Connected:	Indicates PF32 connection status. This will display either: NO – PF32 camera is either not connected, or firmware has not yet been loaded. YES – PF32 camera is connected, and firmware has been loaded.
Model number:	Displays the camera model number.
Serial number:	Displays the camera’s serial number.
Device firmware version:	Displays the version number of the camera’s internal USB firmware.
Bitfile:	Displays the filename of the firmware bit file which has been loaded.
Configuration:	Provides confirmation of the camera’s mode of operation.
SYNC:	Displays the input sync rate.
Duty ratio:	If >50%, input signal is perhaps inverted.

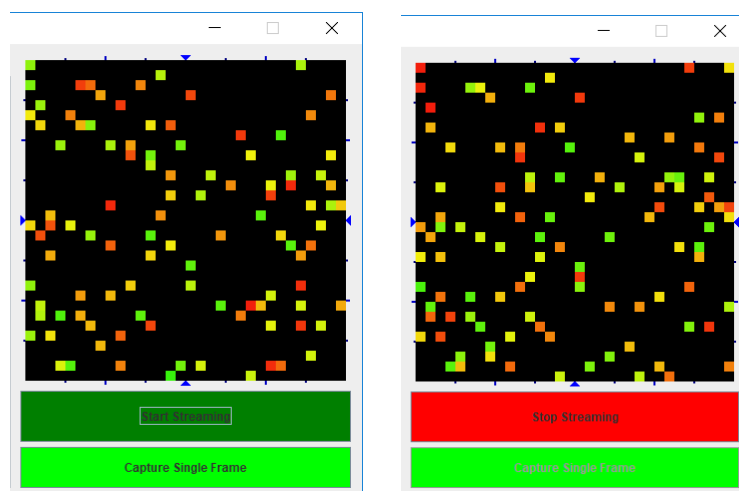
The camera’s operating mode may be quickly set by choosing the desired setting from the “Select mode of operation” drop down box, and pressing the “Configure” button. If successful, the new

configuration will be indicated on the “Configuration” display mentioned above. The following camera modes are available:

Mode	Description
Photon Counting (default)	In this mode, the camera pixels are configured to count the number of detected photons each frame. 7-bit counters are provided, delivering a maximum count of 127 photons per pixel.
TCSPC (laser master)	The camera is configured in Time Correlated Single Photon Counting (TCSPC) mode, where each pixel timestamps the first photon detected during each frame. In this mode, the camera requires a sync signal to be provided from the laser, connected via the SYNC input.
TCSPC (system master)	The camera is configured in TCSPC mode as above, but generates a trigger output signal provided on the TRIG port.

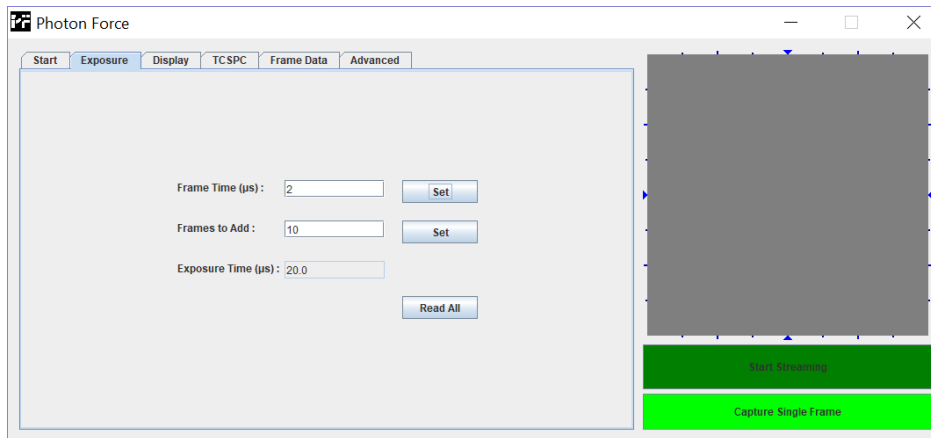
IMAGE DISPLAY AND STREAMING CONTROLS

Once a PF32 camera has been connected, powered on, and configured using the start tab as described above, images may be captured. The streaming controls may be used to capture single frames or stream video rate images from the sensor, for live display in the image display area. Clearly, a PC monitor cannot display the frame rates achievable with the PF32 camera, so this feature should be considered as a useful preview tool, for example when aligning optics.



PF32 Java application image display area and streaming controls, showing idle (left) and streaming in progress (right).

EXPOSURE TAB

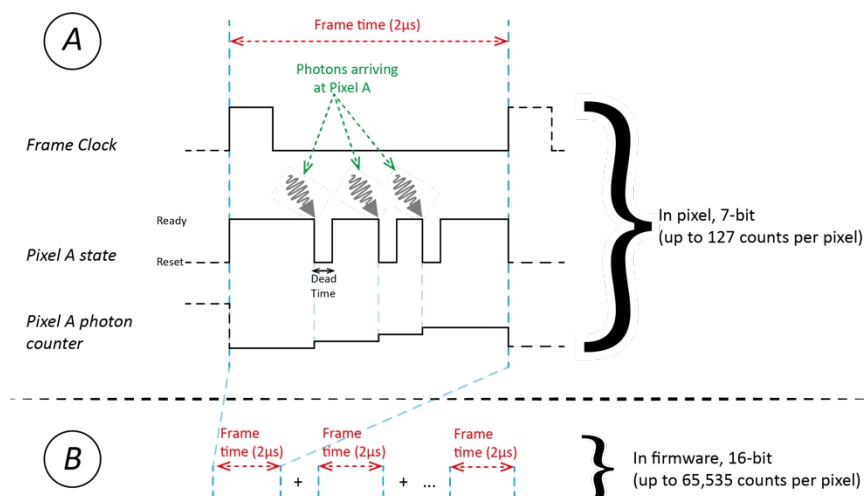


PF32 Java application Exposure tab.

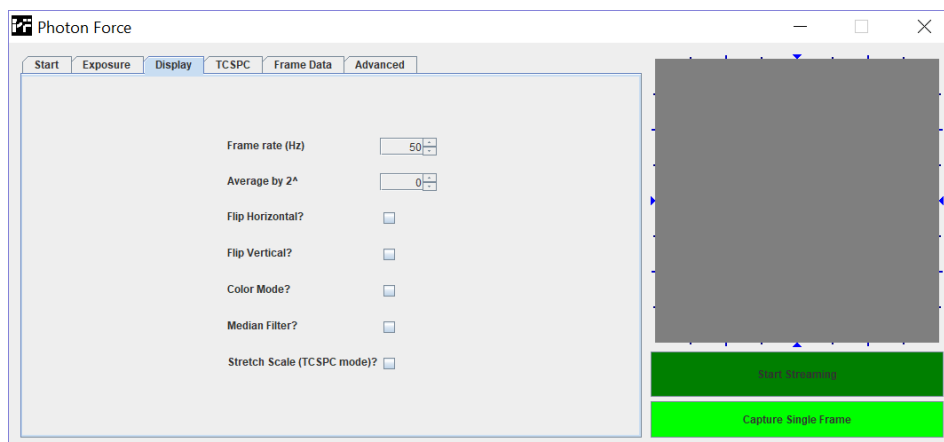
The exposure tab allows the camera's frame time to be set, specified in microseconds (μs). Please note that unless otherwise specified, the camera provides a minimum frame time of $2\mu\text{s}$.

Setting	Description
Frame Time (μs)	Specifies the sensor's frame time in microseconds. Valid values are 2 to $8,000\mu\text{s}$.
Frames to Add	This specifies the number of sensor frames which will be digitally summed by the camera to produce each frame transmitted to the PC. Valid values are 1 to 50,000.
Exposure Time (μs)	Displays the total exposure time for each frame received by the PC, defined as the sensor frame time multiplied by the number of frames to add.

"Frame time" and "Frames to Add" settings are explained in the following figure. Section A demonstrates the measurement procedure within the specified "frame time" to give 7-bit images from the sensor (up to 127 counts per pixel per frame). Care should be taken to ensure that the exposure time and photon flux are such that the in-pixel counter does not overflow. Overflow occurs when more than 127 events are counted per pixel per frame. Section B shows "frames to add" where these individual frames can be summed to give 16-bit images in firmware (up to 65,535 counts per pixel). This firmware summing vastly extends the dynamic range of the sensor images without decreasing the readout speed.



DISPLAY TAB



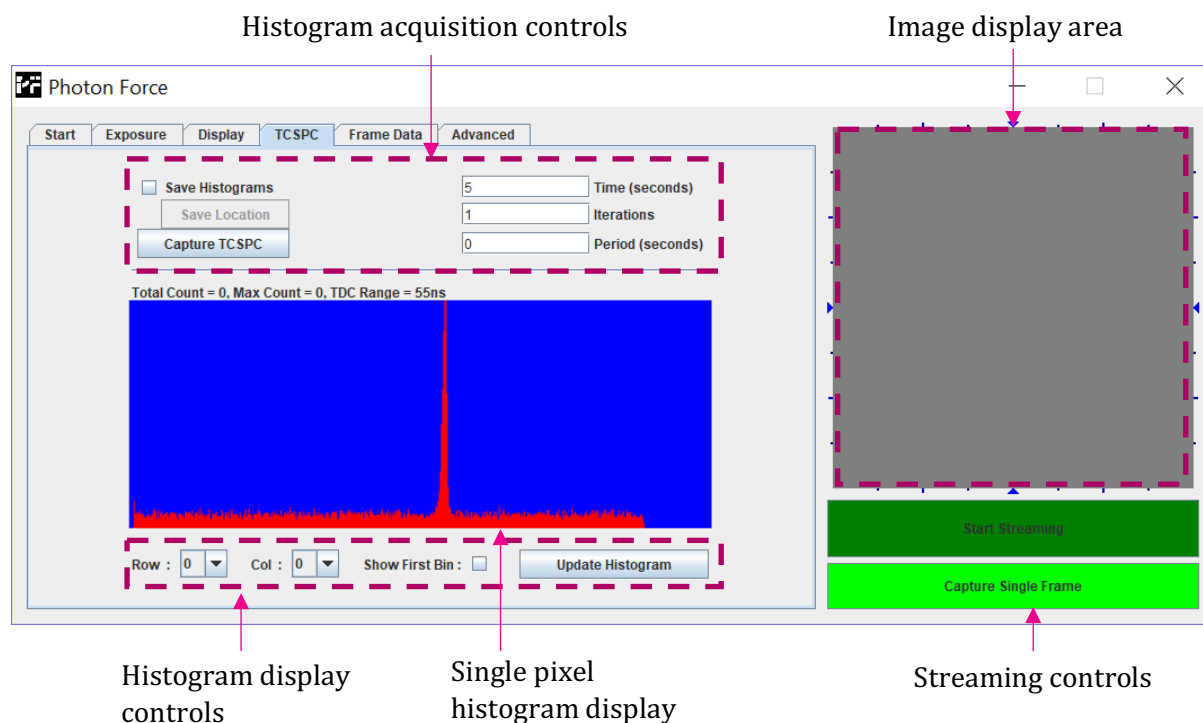
PF32 Java application Display tab.

The display tab provides settings controlling how image data is displayed in the image display area.

Setting	Description
Frame rate (Hz)	Specifies the target frame rate that the application should attempt to display in streaming mode. Note that the refresh rate of most displays is around 60Hz, so streaming faster than this is not useful.
Average by 2^N	This setting instructs the Java application to average a number of sensor frames, specified as 2 ^N , and display the result instead of raw data. This should not be applied in TCSPC mode.
Flip Horizontal	Inverts the displayed image horizontally.
Flip Vertical	Inverts the displayed image vertically.
Color Mode	When checked, enables false colour mode. This is enabled automatically in TCSPC mode.
Median Filter	Applies a noise filter to the displayed image data.
Stretch Scale (TCSPC mode)	

TCSPC TAB

The TCSPC tab provides a basic means of capturing data at high speed from the sensor, and performing software-based histogramming of the data received by each pixel. Note that high speed capture and histogram generation is independent of the image display area and streaming controls discussed previously. The image display area will not update during histogram capture. Also note that if the camera is streaming to the image display area when “Capture TCSPC” is pressed, streaming and image display will pause while histogram capture is in progress, and resume automatically afterwards.



PF32 Java application TCSPC tab.

The following histogram controls are available:

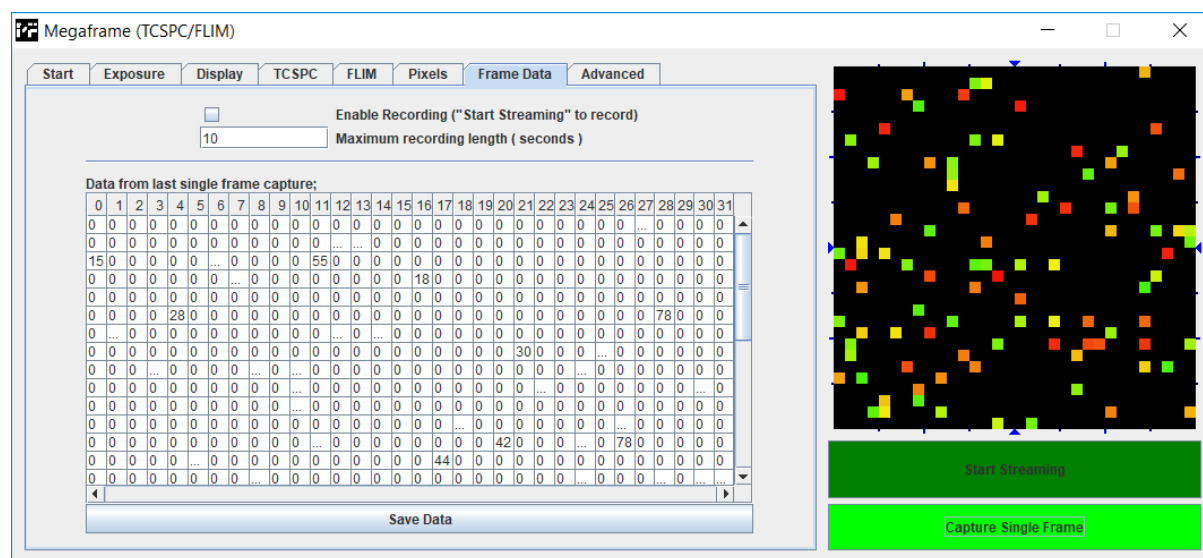
Setting	Description
Save Histograms	If enabled, the software will save the generated histogram data in a CSV format. Alternative file formats may be available on request.
Save Location	If “Save Histograms” is enabled, the user can then choose where the files are to be saved.
Capture TCSPC	The PF32 will start to acquire TCSPC data for the amount of time and iterations as explained below.
Time(seconds)	Histogram update rate, specified in seconds. For example, if set to 1 second, then a number of frames corresponding to 1 second of sensor exposure time will be acquired. Note that data acquisition may take longer than the specified exposure time, depending on the sensor frame rate and USB bus bandwidth.
Iterations	Specifies the number of iterations of data capture to complete. For example, if set to 10, then 10 sets of data of the specified length will be captured.
Period (seconds)	Optionally, a value may be given here to capture the number of iterations within the specified interval (i.e., if time is set to 1, iterations to 10, and period to 10, then a 1-second exposure time will be captured followed by a 9-second pause before the next data capture, repeated 10 times). Note that the period setting will not take effect if set to less than the integration time.

The Histogram tab is able to display histogram data for one pixel at a time. This may be set using the histogram display controls at the bottom of the tab.

Setting	Description
Row	Specifies the row number for which histogram data is displayed.
Col	Specifies the column number for which histogram data is displayed.
Show First Bin	If enabled, histogram bin 0 will be included in the histogram display. Otherwise, bin 0 will not be included. Since bin 0 corresponds to no photon being detected in TCSPC mode, it is not typically useful to include this. At low light levels, the number of no photon (bin 0) counts may vastly exceed the values in other histogram bins.
Update Histogram	When the “Row”, “Col”, or “Show First Bin” settings are changed, the Update Histogram button should be pressed to refresh the histogram display. Note that before a histogram can be displayed, “Calculate Histograms” must be enabled and data must be captured using “Capture TCSPC”.

FRAME DATA TAB

The frame data tab provides a means of inspecting the raw individual pixel values within the GUI. Every time the ‘Capture Single Frame’ streaming control is clicked, this will be updated. Please note that the raw values are not updated during streaming.



PF32 Java application Frame Data tab.

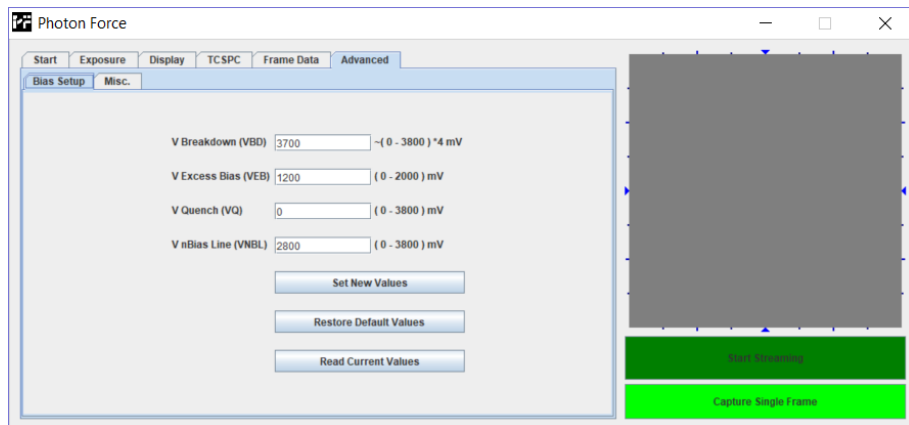
ADVANCED TAB – BIAS SETUP

The Advanced tab contains 2 sub-tabs which provide access to a number of low level features not typically required for customer applications. This tab should be ignored unless specifically advised by Photon Force Ltd. or your distributor.

The bias tab allows the various configuration voltages to be adjusted to tune the performance of your PF32 for specific applications.



Do not adjust the bias voltages without first consulting Photon Force or your authorized distributor. Doing so may result in damage to your camera.



PF32 Java application Bias Setup tab.

SPAD Negative Bias Voltage (VBD)

This is the supply normally set at the reverse breakdown voltage of the detector. Note that the sensor applies a gain of 4, i.e. the default value of 3700 = 14,800mV.

SPAD Excess Bias Voltage (VEB)

The excess bias voltage defines how far beyond the reverse breakdown voltage the SPADs are biased. This is typically set to 1.2V.

SPAD Quench Time Control Voltage (VQUENCH)

VQUENCH allows the sensor's avalanche quenching or recharge time (dead time) to be adjusted. For maximum quenching speed, VQUENCH should be set to the default value of 0V. Increasing this voltage lengthens the recharge time.

TDC Supply Regulation Voltage (VNBIASLINE)

The TDC supply regulation voltage (VNBIASLINE) controls the camera's TDC speed or histogram bin width. The default value of 2800mV corresponds to approximately a 55ps TDC bin width. Decreasing this value results in a slower TDC speed and wider bin width.



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